Census Income Take-Home Report

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1. Executive Summary

This submission solves two connected tasks on the Census Income dataset: weighted binary classification of whether a record belongs to the income group above $50K, and interpretable customer segmentation for marketing activation. The final workflow explicitly addresses class imbalance, preserves structural categories such as `?` and `Not in universe`, and uses survey weights for evaluation and segment profiling.

The classification comparison now includes not only ROC and PR curves, but also threshold policy, calibration, score separation, and bootstrap uncertainty. On the held-out test split, XGBoost attains the highest weighted ROC-AUC at 0.9555, while LightGBM delivers the best weighted PR-AUC at 0.6845 and the best thresholded weighted F1 among the deployed operating points at 0.6137. LightGBM is also the best calibrated model, with weighted expected calibration error 0.0029 versus 0.0036 for XGBoost and 0.1629 for FT-Transformer. For that reason, LightGBM remains the recommended production classifier.

The segmentation section now compares the pure unsupervised baseline (`V1`) and the classifier-informed extension (`V2`) in the same family of figures. Version 1 yields the better cluster-quality trade-off for the main submission, with $k = 5$, silhouette 0.381, and composite score 1.722. Version 2 is more stable, with pairwise agreement 0.975, and it splits the high-income population into several finer premium-oriented subsegments. The recommended client-facing story is therefore: use LightGBM for propensity scoring, keep Version 1 as the main segmentation result, and use Version 2 as an advanced activation layer when finer premium targeting is needed.

2. Problem Framing And Data Constraints

The take-home requires a report that is both technically credible and usable by a business client. The dataset contains 40 demographic and employment variables, one survey-weight column, and one binary income label. Four constraints are most important:

- **Class imbalance.** The positive class is much smaller than the negative class, so PR-AUC, weighted log loss, threshold tuning, and ranking quality matter more than raw accuracy.
- **Structural missingness.** Tokens such as `?` and `Not in universe` carry signal and should not be dropped or blindly imputed away.
- **Mixed-type, high-cardinality tabular features.** This motivates model-specific handling across CatBoost, XGBoost, LightGBM, and FT-Transformer.
- **Survey weights.** The `weight` column is excluded from the feature space, but used in training where supported, in all reported evaluation metrics, and in segment summaries.

The final splits contain 119,713 training rows, 39,905 validation rows, and 39,905 test rows. A persistent row identifier is used throughout to prevent leakage and to keep segmentation artifacts joinable.

2.1. Exploratory Data Analysis

Figure 1 gives one compact data-exploration checkpoint. The positive class is small in both row count and weighted population share, the survey weights have a broad distribution, and the raw age / weeks-worked differences already suggest why income prediction is learnable in this dataset.

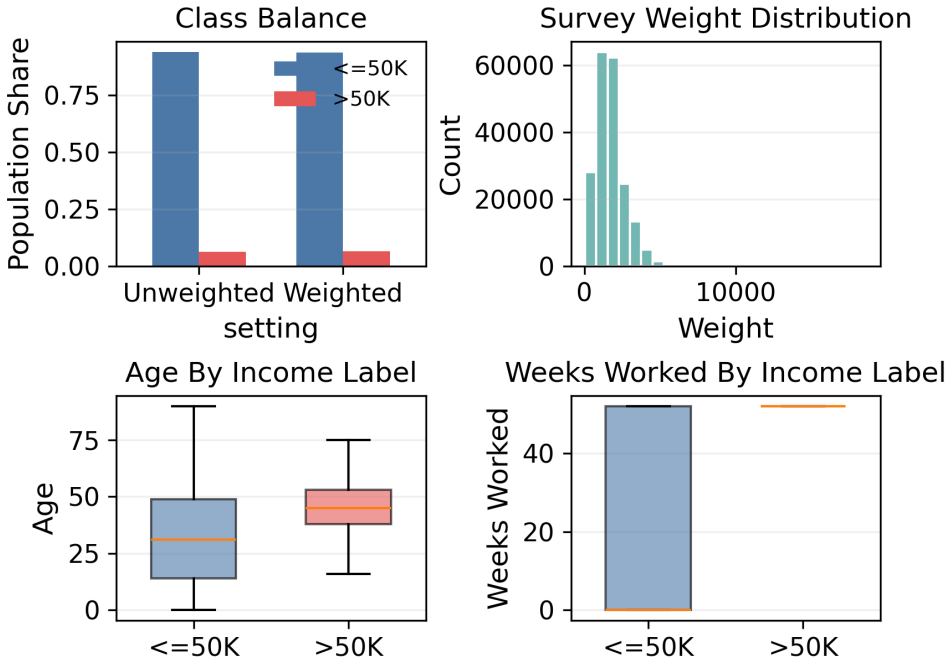


Figure 1: Exploratory data analysis overview: class balance, weight distribution, age by income label, and weeks worked by income label.

3. Modeling Approach

3.1. Classification

The classification stack compares CatBoost, XGBoost, LightGBM, and FT-Transformer. Exported prediction artifacts keep a logit-first design:

$$\texttt{logit} \rightarrow \texttt{probability = sigmoid(logit)} \rightarrow \texttt{predicted_label}$$

This is important because the same score supports ranking, calibration, and business threshold changes without rerunning the full model.

Preprocessing and training decisions are aligned with the take-home brief. Structural categories such as `?` and `Not in universe` are preserved as categories rather than collapsed into ordinary missingness. The pipeline uses model-specific handling for mixed-type inputs, persistent row identifiers for fixed train/validation/test splits, and sample weights in training where the estimator supports them. All final model comparisons are reported on the hold-out test split, while the operating threshold is tuned on validation only. This separation is important because the deployed decision rule should be selected before looking at final test performance.

3.2. Segmentation

Two segmentation pipelines are retained:

- **Version 1:** pure raw-feature unsupervised segmentation using the engineered demographic and employment feature set only.
- **Version 2:** classifier-informed segmentation that augments the same feature set with the model-derived income propensity score.

Version 1 is built from a curated demographic and employment feature subset that includes numeric variables such as age, weeks worked, veterans benefits, and number of persons worked for employer; engineered binary asset indicators for capital gains, losses, and dividends; and banded variables such as age band and weeks-worked band. Numeric columns are median-imputed and scaled, categorical columns are most-frequent-imputed and one-hot encoded, and the resulting feature space is used as the input to MiniBatchKMeans. Version 2 reuses exactly this V1 feature space and adds the model-derived income

propensity score as an auxiliary signal, which shifts the clustering geometry toward finer premium-oriented splits. The true income label is still excluded, so V2 is classifier-informed but not supervised clustering with respect to the target label.

Cluster count is not chosen by one metric alone. The pipeline searches $k \in \{3, 4, 5, 6, 7, 8\}$ and ranks candidates with the internal score 'silhouette + 1/(1+DB) + log(1+CH)/10 - 0.5 * 1(min cluster share < 1%)'. This rewards separation, penalizes overlap, adds a tempered compactness term, and subtracts a fixed penalty for tiny clusters. Stability is then used as a post-selection robustness check rather than a direct ingredient in the composite score, which keeps the k selection rule transparent.

4. Classification Results

4.1. Cross-Validation And Hold-Out Performance

Table 1 summarizes mean weighted cross-validation performance. LightGBM and XGBoost clearly outperform CatBoost and FT-Transformer, with nearly identical ROC behavior and a small PR/log-loss edge for LightGBM.

Table 1: Mean weighted cross-validation metrics.

Model	ROC-AUC	PR-AUC	Weighted Log Loss
CatBoost	0.943	0.627	0.127
XGBoost	0.952	0.682	0.116
LightGBM	0.952	0.685	0.115
FT-Transformer	0.937	0.563	0.339

Table 2 uses the exported test predictions. XGBoost gives the best ROC-AUC, but LightGBM remains the preferred final model because it improves PR-AUC and operating F1 while also achieving the lowest weighted log loss. This matters more for an imbalanced targeting problem than a marginal ROC gain alone.

Table 2: Held-out weighted test metrics. Operating F1 uses each model's deployed threshold (0.38 for LightGBM, 0.50 for XGBoost, 0.82 for FT-Transformer).

Model	ROC-AUC	PR-AUC	Weighted Log Loss	Operating F1
XGBoost	0.9555	0.6799	0.1141	0.5746
LightGBM	0.9549	0.6845	0.1139	0.6137
FT-Transformer	0.9413	0.5775	0.3115	0.5482

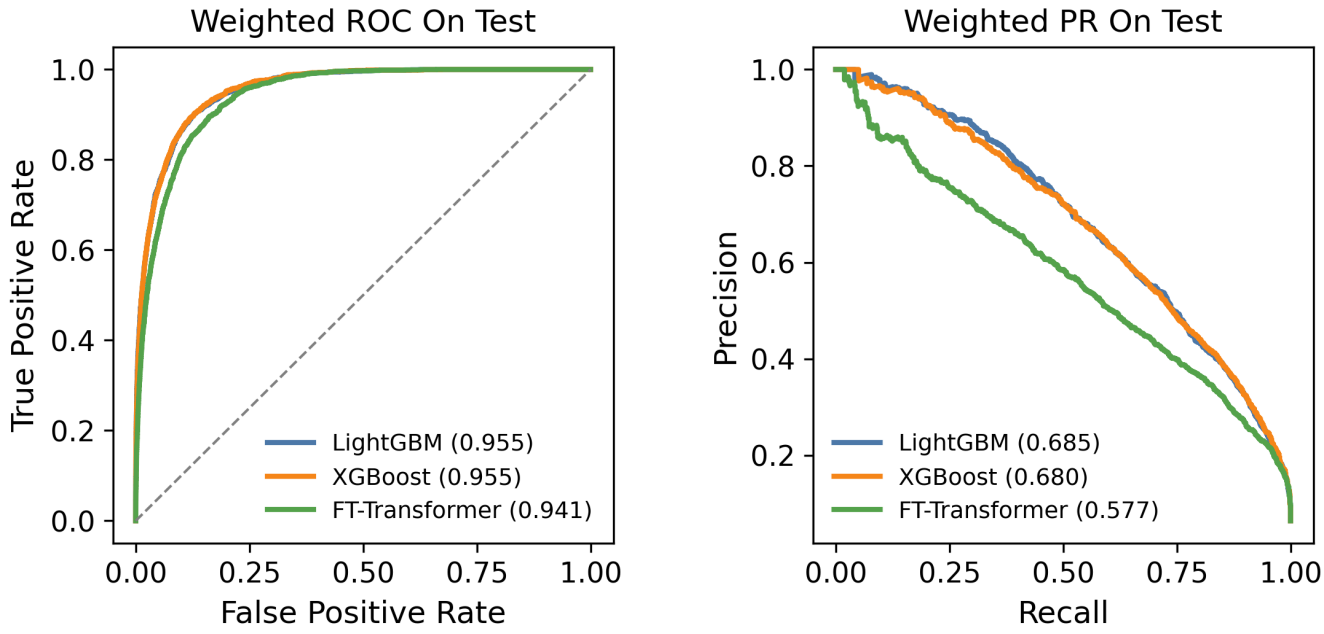
Figures 2 and 3 show why the model-selection story is stronger than a single metric table. XGBoost and LightGBM are almost tied on ROC, but LightGBM keeps slightly better precision in the actionable recall region and is materially better calibrated. The FT-Transformer baseline is not only weaker in discrimination but also much less well calibrated.

4.2. Why Boosted Trees Won Here

I do not interpret the LightGBM result as a generic statement that deep tabular models are weak. I interpret it as a dataset-regime judgment. This problem is a mixed-type census table with strong categorical structure, moderate feature engineering leverage, and clear benefit from well-calibrated ranking scores; in that regime, boosted trees are often the safer and stronger choice. The evidence here is consistent across metrics: LightGBM and XGBoost beat FT-Transformer on PR-AUC, weighted log loss, deployed operating F1, and calibration. FT-Transformer was retained deliberately as a neural tabular benchmark to test whether learned representations could outperform gradient boosting on this feature regime, but it did not. In production I would therefore treat FT-Transformer as an informative benchmark, not as a competitive deployment candidate for this dataset.

4.3. Threshold Policy, Calibration, And Uncertainty

The validation threshold sweep for LightGBM converts the threshold from a magic number into a policy decision. Figure 3 shows that weighted F1 peaks around 0.38, exactly matching the saved threshold artifact.



(a) Weighted ROC on test.

(b) Weighted PR on test.

Figure 2: Discrimination quality of the three exported classifiers.

At that point, the targeted weighted population share is about 5.8% on the validation split. This is a useful operating point for selective outreach: precision remains high while recall is still meaningfully above the level that a 0.50 threshold would produce.

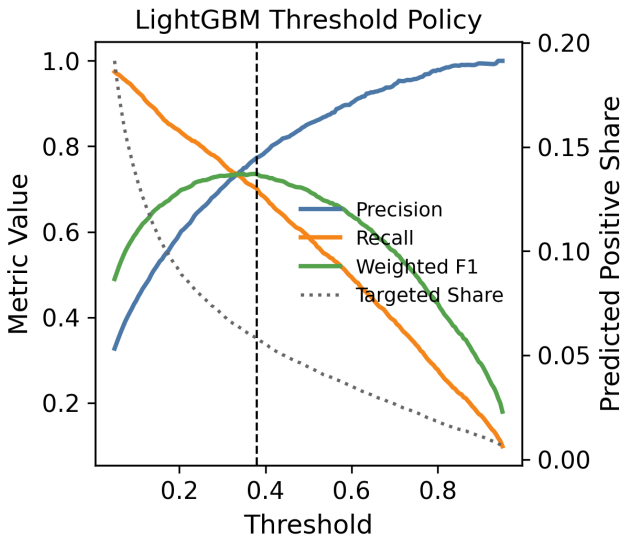
The same figure also reports calibration. LightGBM has weighted ECE 0.0029 and XGBoost has weighted ECE 0.0036, indicating that both tree ensembles are well-behaved probability models. FT-Transformer has ECE 0.1629 and therefore is not reliable enough to use as a direct calibrated confidence score without substantial post-processing.

4.4. Business Decision Framework

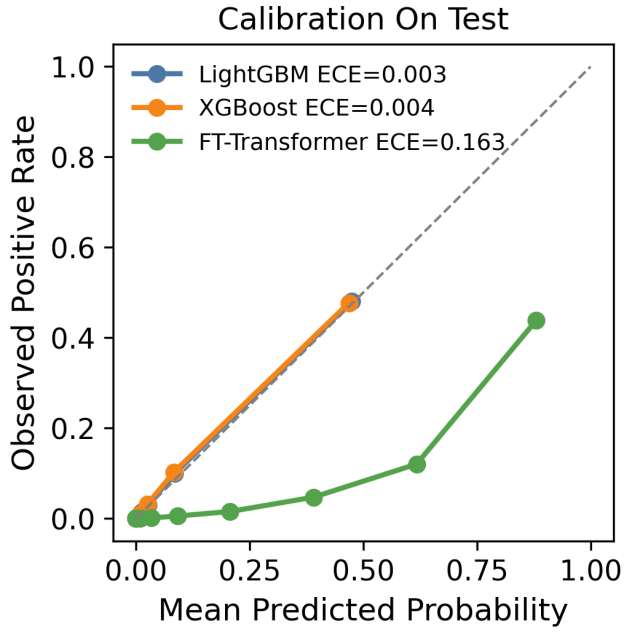
The preferred operating objective is **selective outreach for premium margin efficiency**, not maximal audience reach. That is why precision matters more than raw recall at deployment time: a marketing client typically pays for every premium impression, contact, or offer, so false positives create direct budget waste and channel fatigue. At the selected threshold of 0.38, the model targets about 5.8% of weighted population on validation, which is operationally consistent with a focused premium campaign rather than a broad awareness push. The ranking view tells the same story on test: the top 5% scored weighted population reaches 68.8% observed positive rate, or about 10.8x lift over the 6.4% weighted base rate; the top 10% still reaches 48.5%, or about 7.6x lift. I recommend treating this threshold as a policy choice, not merely a validation artifact. It is appropriate when the client has limited high-value offer inventory and wants stronger expected value per contact rather than maximum coverage.

Figure 4 adds two more pieces of evidence. The score distribution plot shows that the LightGBM probability mass for the positive class is shifted clearly to the right, though with some overlap around the decision region as expected in census income data. The three-model performance comparison with bootstrap 95% intervals shows that the ranking among models is not a fluke: LightGBM and XGBoost overlap strongly on ROC, but LightGBM has the strongest thresholded F1 profile among the deployed operating thresholds.

The bootstrap intervals are narrow enough to support a stable conclusion. For LightGBM, the 95% bootstrap interval is approximately [0.9515, 0.9585] for ROC-AUC, [0.6658, 0.7009] for PR-AUC, and [0.5956, 0.6317] for operating F1. Those intervals support the decision to treat the model as production-ready.

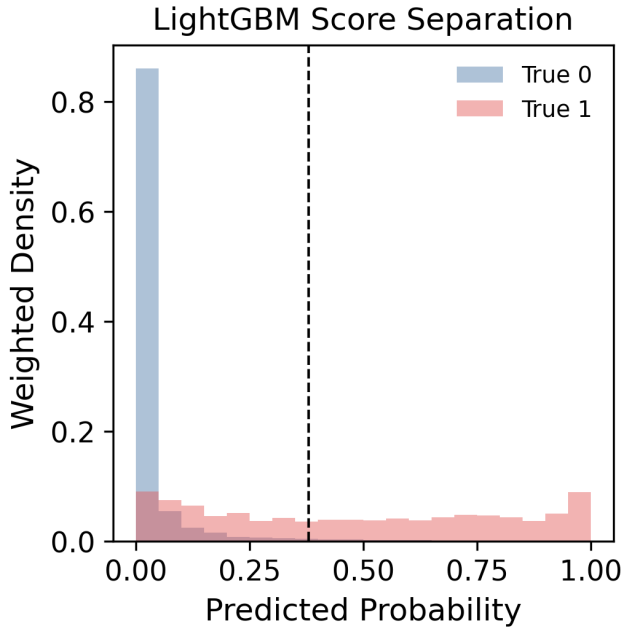


(a) Validation threshold policy for LightGBM.

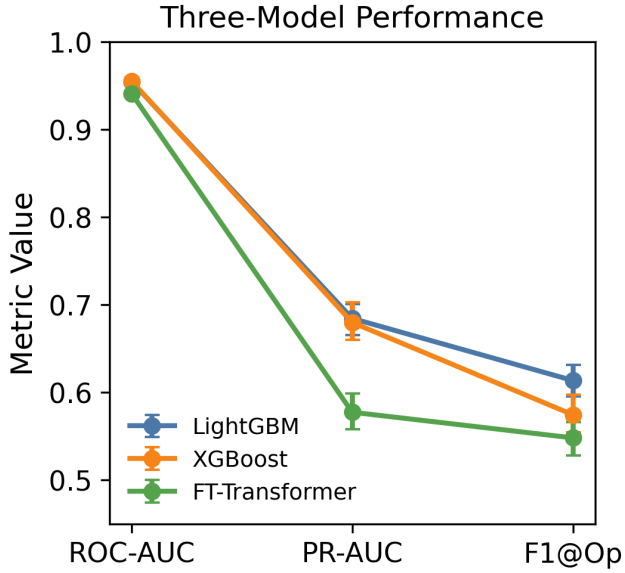


(b) Weighted calibration comparison on test.

Figure 3: Threshold choice and score calibration.

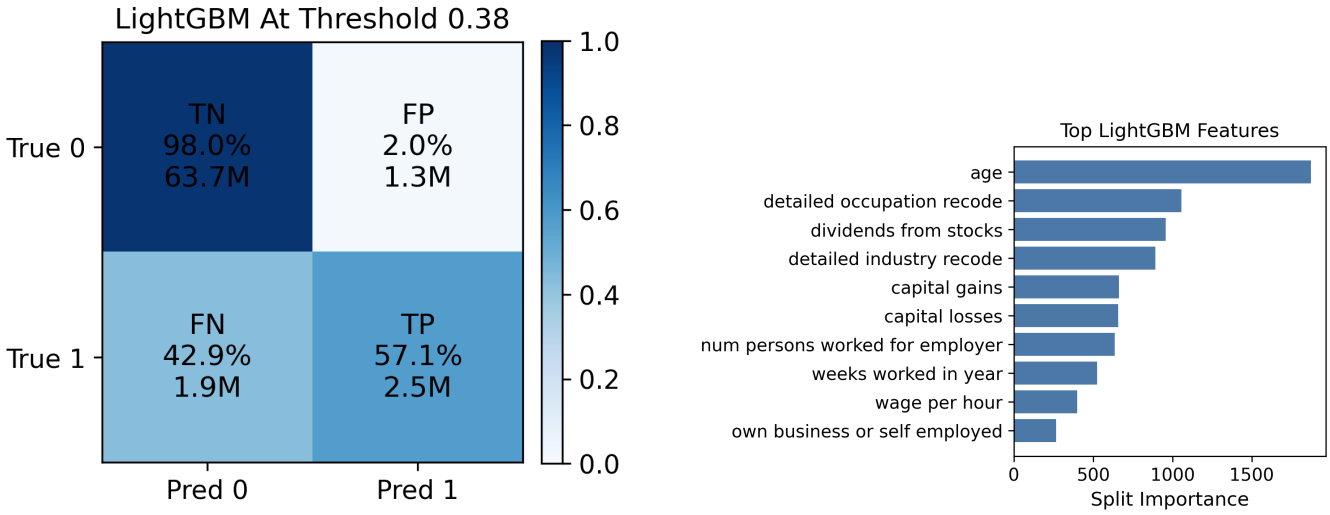


(a) LightGBM score separation by true class.



(b) Three-model performance with bootstrap CI.

Figure 4: Confidence structure and uncertainty-aware model comparison.



(a) Weighted confusion matrix at threshold 0.38.

(b) Top LightGBM split-based importances.

Figure 5: Operating behavior and model explanation for the selected classifier.

Figure 5 closes the classification argument. The confusion matrix shows that the tuned threshold still controls false positives while substantially recovering positives. The feature-importance summary matches domain intuition: age, occupation, dividends, industry, capital gains and losses, and weeks worked all remain strong predictors of higher income. I treat this split-based importance plot as directional evidence rather than a final causal explanation. If I were extending the project further, the next explanation upgrade would be permutation importance or a compact SHAP summary to test whether the same feature ranking remains stable under a stronger model-agnostic lens.

5. Segmentation Results

5.1. Version 1 Vs Version 2

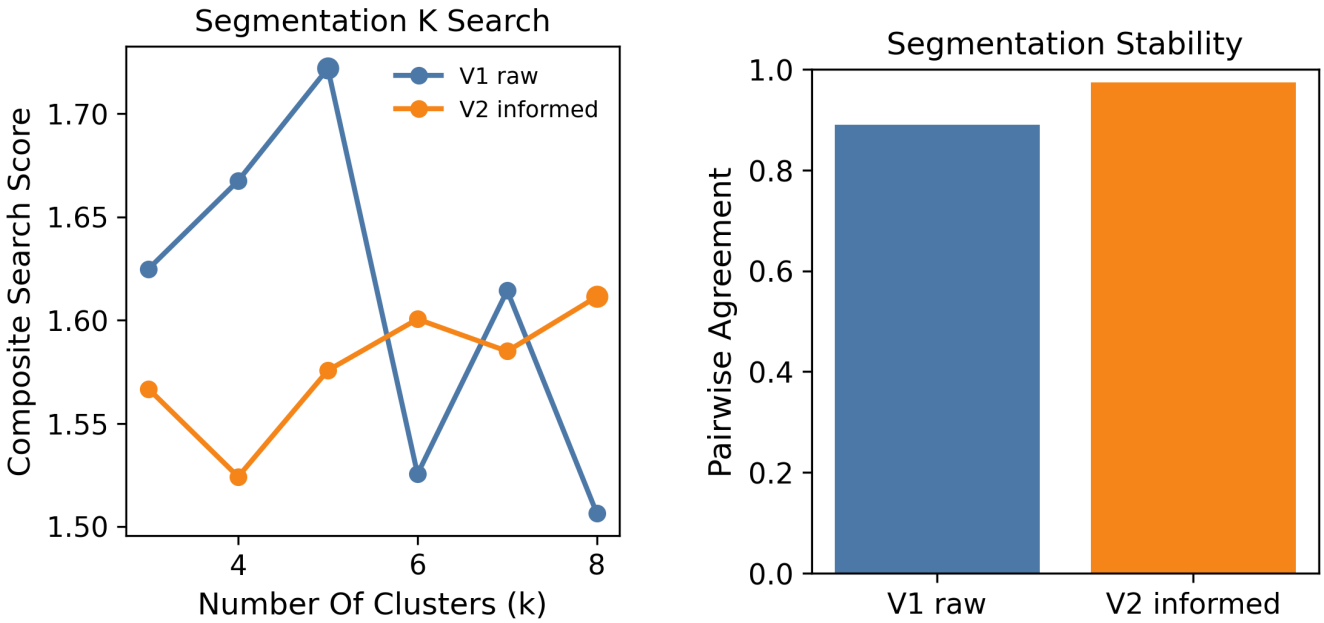
The segmentation analysis is now organized as an explicit comparison between the pure unsupervised baseline and the classifier-informed extension. This makes the trade-off transparent: Version 1 is cleaner and more interpretable as a primary submission artifact, while Version 2 is useful for finer-grained actionability.

Figure 6 and Table 3 show the main pattern. Version 1 selects $k = 5$ and achieves the better composite score and silhouette, while Version 2 selects $k = 8$ and achieves markedly higher stability. This is a sensible outcome: adding the classifier score creates a more repeatable structure, but it also pushes the space toward finer, more score-driven premium splits.

I would report V1 as the primary segmentation in the take-home submission and keep V2 as the activatable downstream variant. V1 gives the cleaner client narrative because every segment can be explained directly from census-style demographic and labor variables. V2 is more operationally detailed because it uses the classifier score to separate the affluent tail into narrower premium subsegments, but that also makes it more dependent on the quality and stability of the supervised model. If the client wants a simple operating workflow, I would deploy V1 only and not expose V2. If the activation team can support finer premium subsegments, I would use V2 only downstream after the primary scoring decision has already been made.

Table 3: Segmentation pipeline comparison.

Scheme	k	Best Score	Silhouette	Stability
V1 raw-feature unsupervised	5	1.722	0.381	0.891
V2 classifier-informed	8	1.611	0.337	0.975



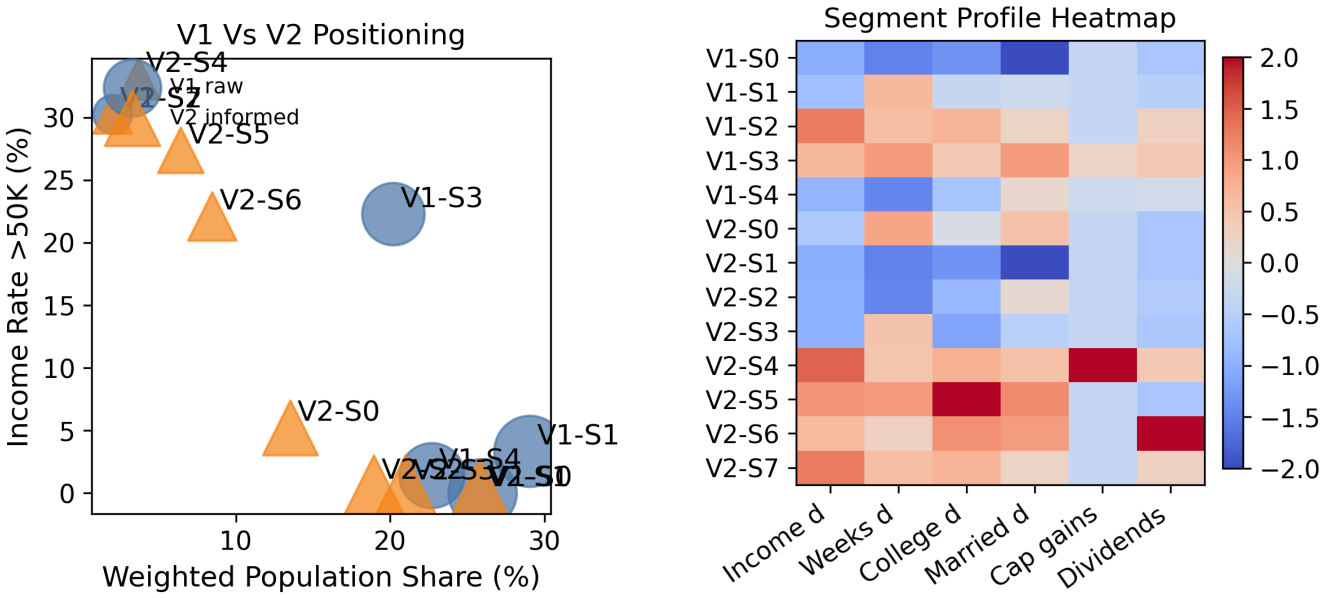
(a) K-search score for V1 and V2.

(b) Stability comparison.

Figure 6: Model-selection evidence for the two segmentation variants.

5.2. How The Segment Structures Differ

The next set of figures replaces text-only persona descriptions with auditable evidence. Figure 7 shows the same V1 and V2 segments in a common positioning chart and a common profile heatmap. Version 1 has one small, college-leaning professional niche and one broader married full-year working segment. Version 2 preserves those low-income structural groups, but it splits the high-income region into several more targeted segments. In other words, the classifier-informed pipeline does not invent a new market structure from nothing; it refines the affluent tail.



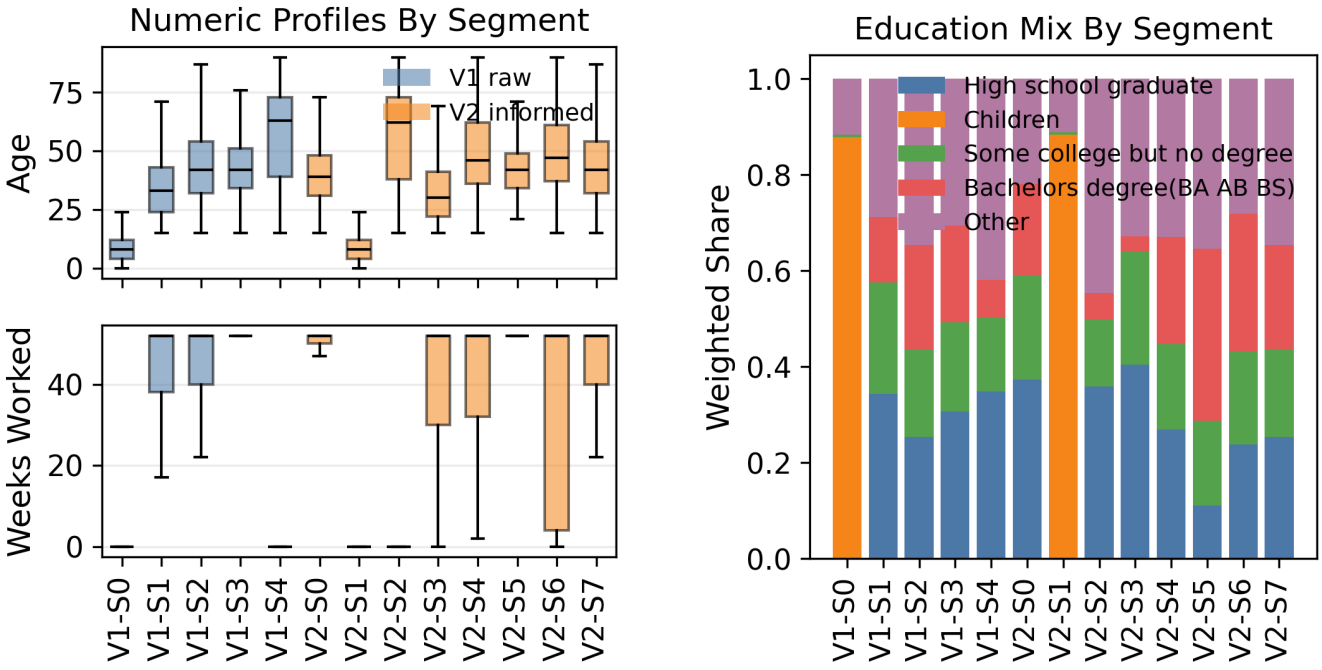
(a) Positioning of V1 and V2 segments.

(b) Joint standardized profile heatmap.

Figure 7: Direct comparison of V1 and classifier-informed V2 segment structure.

The heatmap is especially informative. The V1 high-value segments already have positive deltas on income rate, weeks worked, college attainment, and marriage. Version 2 then separates that high-value

population into multiple subsegments with different asset and education signatures, including one capital-gains-heavy cluster and several college-dense mid-career clusters. That is why V2 is useful as an advanced marketing overlay even though V1 remains the better main report choice.



(a) Numeric differences by segment.

(b) Education mix by segment.

Figure 8: Distributional and categorical evidence for segment distinctness.

Figure 8 provides more rigorous support for the personas. The numeric boxplots show that age and labor intensity are not only different in their means but also in their distributional spread. The education composition chart shows that the classifier-informed high-income segments become more Bachelor's-heavy and more differentiated than in Version 1, which is exactly the kind of refinement expected when the clustering algorithm is allowed to use a propensity score.

From a retail-marketing mechanics perspective, these differences affect product tier, promotion aggressiveness, contact frequency, and expected ROI per impression. Higher-income, college-dense, labor-attached segments justify premium assortment, lower discount depth, and higher-touch channels because the expected margin per contact is larger. Lower-propensity or structurally non-working groups should receive lower-cost awareness messaging, softer promotions, and lower contact frequency because repeated premium offers are more likely to create waste than conversion. This is why the segmentation layer is useful: it changes not only who gets contacted, but also what gets offered and through which channel.

5.3. Version 1 Personas For The Main Submission

Even after adding the V2 comparison, Version 1 remains the best main-report segmentation because it is easier to explain to a client without relying on a supervised score. Table 4 keeps the Version 1 persona layer in a concise format.

The key business interpretation is straightforward. Segment S2 (professional niche) is the smallest but highest-value niche for premium campaigns. Segment S3 (full-year upper-mid) is the scalable middle-to-upper tier audience. The quantitative reason is clear in Table 5: S2 has the highest rate-based efficiency at 30.2% income incidence, but S3 carries about 7.35x as much expected positive mass because its weighted share is much larger while its income rate remains high at 22.3%. Segments S0 and S4 should not absorb premium budget. Segment S1 is a broad mainstream audience suited to value-oriented but still personalized communication.

I recommend the following deployment priorities. If premium campaign budget is limited, prioritize

Table 4: Version 1 evidence A: segment personas and activation guidance.

ID	Persona	Share	Income Rate	Suggested Action
S0	Youth / dependent households	26.0%	0.0%	De-prioritize premium targeting; use low-cost or family-oriented messaging.
S1	Mainstream employed clerical households	29.1%	3.3%	Use value messaging, practical bundles, and moderate-contact campaigns.
S2	College-leaning professional niche	2.0%	30.2%	Target with premium products, cross-sell, and high expected-value offers.
S3	Married full-year upper-mid households	20.2%	22.3%	Position upper-mid-tier offers around reliability, family value, and productivity.
S4	Older non-working households	22.7%	1.4%	Use low-frequency retention and age-appropriate service messaging.

Table 5: Version 1 evidence B: absolute profile summary.

Seg	Share	Income	Weeks	College+	Married	Cap gains+
S0	26.0%	0.0%	0.1	0.0%	0.0%	0.0%
S1	29.1%	3.3%	43.0	18.4%	46.3%	0.0%
S2	2.0%	30.2%	41.5	37.4%	57.9%	0.0%
S3	20.2%	22.3%	49.4	32.0%	75.8%	15.0%
S4	22.7%	1.4%	1.0	11.3%	55.9%	3.0%

S2 first for margin efficiency and S3 second for scalable upper-mid-tier conversion. If the goal is scale-maximizing expansion, use S3 first because it is materially larger while still retaining strong income propensity. If the goal is low-cost awareness, S1 is the best broad audience and S0/S4 should only be touched with inexpensive channels or family/lifecycle relevant messaging. This trade-off is acceptable because the segment structure cleanly separates expected value from audience size rather than forcing one compromise segment to do both jobs.

6. Final Recommendation

The final recommended stack is:

- **Primary classifier:** LightGBM. It combines the best PR-AUC, best weighted log loss, best operating F1 among the deployed models, and the best calibration profile.
- **Primary segmentation:** Version 1 raw-feature unsupervised segmentation. It provides the cleanest methodological story and the strongest balance of interpretability and cluster quality.
- **Advanced segmentation overlay:** Version 2 classifier-informed segmentation. It should be used when the client wants finer premium-audience splits and is comfortable with a segmentation layer informed by the classifier score.

The preferred deployment choice is to first rank the population with the LightGBM score, then personalize campaign content and product tier using the Version 1 segments. If a downstream activation team wants more refined high-income subsegments, the Version 2 assignments can be used as a secondary overlay.

I would not use classifier-only targeting as the entire operating policy, even though the classifier is strong. The score is excellent for ranking propensity, but it does not by itself decide product tier, promotion aggressiveness, channel choice, or contact frequency. That independent value is exactly why segmentation should remain in the workflow. I also would not launch V2 if the activation team cannot operationalize finer premium subsegments, because in that setting the extra complexity would add governance cost without enough incremental value.

In production I would deploy LightGBM first, expose V1 segmentation in the client-facing workflow, and keep V2 behind the scenes as an advanced activation option. I would monitor calibration drift, segment drift, and sparse-category instability over time, because those are the most likely ways the current decision logic could degrade after launch. I would also explicitly guard against over-targeting premium offers to structurally low-propensity groups such as S0 and S4, because that failure mode creates budget waste while also weakening the credibility of the targeting system.

7. References

- LightGBM documentation: Microsoft LightGBM.
- XGBoost documentation: XGBoost project documentation.
- CatBoost documentation: Yandex CatBoost documentation.
- FT-Transformer benchmark reference: Gorishniy et al., “Revisiting Deep Learning Models for Tabular Data,” NeurIPS 2021.
- Calibration reference: Niculescu-Mizil and Caruana, “Predicting Good Probabilities with Supervised Learning,” ICML 2005.
- Clustering metrics reference: scikit-learn documentation for silhouette, Davies-Bouldin, and Calinski-Harabasz indices.

8. Reproducibility Notes

All report figures are generated directly from saved project artifacts by:

```
python plot/scripts/generate_report_figures.py
```

The script writes all report figures into `plot/fig/`. The report source file is `report.tex` at the repository root, on the same level as the project `README.md`. If compiled with XeLaTeX or LuaLaTeX, the main font is Times New Roman. If compiled with pdfLaTeX, the document falls back to a portable Times-style package.